

The Dimension of the Two-Adic Hecke Algebra at Odd Level

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Abstract

For every odd positive integer N , every irreducible component of the full two-adic Hecke algebra of level $\Gamma_1(N)$ has Krull dimension four. This proves the $p = 2$ case of Emerton's dimension conjecture, including all residual components.

1 Introduction

The congruences between classical modular forms of different weights assemble their Hecke eigenvalues into a single p -adic algebra. Its spectrum is larger than an individual family of eigenforms: many such families can pass through the same classical point. The dimension of this spectrum measures how much variation is present when no slope or residual irreducibility condition is imposed.

Fix an odd positive integer N . For $i \geq 1$, let $M_i(N)$ be the complex vector space of modular forms of weight i and level $\Gamma_1(N)$, including Eisenstein series. For every prime $\ell \nmid N$, write T_ℓ for the usual Hecke operator and let S_ℓ act on $M_i(N)$ as $\ell^{i-2}\langle\ell\rangle$, where $\langle\ell\rangle$ is the diamond operator. Let

$$T_{\leq k}^{(2)}(N) \subseteq \text{End}_{\mathbb{C}}\left(\bigoplus_{i=1}^k M_i(N)\right)$$

be the $\mathbb{Z}$ -algebra generated by T_ℓ and ℓS_ℓ for $\ell \nmid 2N$. Set

$$A_k = \mathbb{Z}_2 \otimes_{\mathbb{Z}} T_{\leq k}^{(2)}(N), \quad A = T_2(N) = \varprojlim_k A_k, \quad (1.1)$$

with transition maps given by restriction. The topology on A is the inverse-limit topology, with the 2-adic topology on each A_k . All dimensions below are Krull dimensions.

Theorem 1.1. *For every odd positive integer N , every irreducible component of $\text{Spec } T_2(N)$ has dimension exactly 4.*

This is the $p = 2$ case of Emerton's Conjecture 2.9 [8]. The algebra in (1.1) retains all weights, all Eisenstein systems, and all residual systems of eigenvalues. In particular, the theorem does not require a residual Galois representation to be irreducible, nonscalar, or distinguished at 2.

History and the main input

Hida’s ordinary families [10] and the eigencurve of Coleman and Mazur [5] give fundamental constructions of p -adic families of modular eigenforms. Gouvêa and Mazur’s infinite fern [9] explains how families intersect and produce higher-dimensional Zariski closures. Emerton’s account [8] formulates the dimension question for the full prime-to- Np Hecke algebra, without restricting to one residual deformation problem. His Corollary 2.28 gives the lower bound needed here: every irreducible component has dimension at least 4.

The complementary upper bound is a deformation-theoretic question. Mazur’s deformation theory [14] relates infinitesimal deformations of a Galois representation to its adjoint cohomology. For automorphic representations, the finite local conditions of Bloch and Kato [2] isolate a Selmer group within this cohomology. Under suitable hypotheses, Kisin [13] proved geometric adjoint Selmer vanishing and applied it to local deformation and eigencurve geometry. Allen [1] established adjoint Selmer vanishing under residual hypotheses and related local genericity to unobstructed deformations. Newton and Thorne [16] subsequently proved the vanishing theorem used here without the residual restrictions that would obstruct its application to all components at 2.

We use their Theorem 5.4 for classical cuspidal forms over $\mathbb{Q}$. The theorem includes $p = 2$ and also includes a CM form provided its CM field is not contained in the cyclotomic 2-power extension. Odd level guarantees this last condition. The result concerns the *full* adjoint representation, of dimension four, which is the one required when the determinant is allowed to vary.

Method and organization

The proof works at characteristic-zero points, even on components whose residual systems are reducible or scalar. There are three steps. First, the integral algebra is a finite product of complete Noetherian local rings, and its classical points are Zariski dense. At 2 we justify Noetherianity using Chenevier’s determinant laws [4], avoiding division by 2 in the integral pseudorepresentation. Second, the Eisenstein locus has dimension at most 2. Together with Emerton’s lower bound, this supplies a classical cuspidal point of weight at least 3 on every component. Third, the tangent space at such a point has dimension at most 3: the Newton–Thorne theorem annihilates the global finite Selmer group, and the only nonzero local quotient has dimension 3 at 2. A dimension formula then adds the one-dimensional arithmetic quotient.

Two passages in this argument are useful beyond the immediate application. Section 3 constructs the Eisenstein interpolation maps directly from the inverse-limit Hecke algebra by a compact graph argument. Section 4 gives a characteristic-zero tangent bound for a compact Noetherian algebra carrying a two-dimensional determinant. In particular, it proves the continuity of the derivations that occur in its algebraic tangent space. Neither passage assumes residual irreducibility.

We supply these arguments in detail. The deep external inputs are the classical attachment and local compatibility of Galois representations, Emerton’s lower bound, Chenevier’s determinant theorems, and Newton–Thorne’s Selmer vanishing. Their relevant forms and hypotheses are stated where they are used.

2 The integral algebra and its Galois determinant

Our first goal is to put the inverse limit (1.1) in the category of Noetherian rings while retaining all of its classical systems. Write S for the set of finite primes dividing $2N$, and let $G = G_{\mathbb{Q},S}$ be the Galois group of the maximal extension of $\mathbb{Q}$ unramified at finite primes outside S . Ramification at infinity is allowed. All Galois cohomology will be continuous cohomology.

2.1 Finite stages and classical systems

We use the classical rationality, integrality, and simultaneous semisimplicity of the prime-to-level Hecke action; a convenient statement including weight one and Eisenstein forms is [8, Proposition 1.12 and Remark 1.13]. For cusp forms, old copies have the same good-prime eigenvalues as their associated newforms. On the Eisenstein subspace the eigenvalues are sums and products of Dirichlet character values and powers of primes.

Consequently, for every nonzero finite stage there is an embedding

$$A_k \hookrightarrow \prod_{j=1}^{m_k} \mathcal{O}_{E_{k,j}}, \quad (2.1)$$

where each $E_{k,j}/\mathbb{Q}_2$ is finite and the coordinate maps are classical eigenvalue systems, with all necessary 2-adic embeddings included. Indeed, before tensoring with $\mathbb{Z}_2$ the algebra embeds into a finite product of rings of integers of number fields. It is therefore finite and torsion-free over $\mathbb{Z}$; flatness of $\mathbb{Z}_2$ preserves the embedding. Thus A_k is finite free over $\mathbb{Z}_2$, and its image in (2.1) is closed. Zero initial stages, if present, can be discarded.

A *classical point* of $\mathrm{Spec} A$ means the kernel of a classical eigenvalue map $A \rightarrow \mathcal{O}_E$ obtained from some finite stage. The following consequences will be used repeatedly.

Lemma 2.1. *The ring A is compact, Hausdorff, and reduced. Its classical points are Zariski dense, and the operators $T_\ell, \ell S_\ell$, $\ell \nmid 2N$, topologically generate it over $\mathbb{Z}_2$.*

Proof. The restriction maps between finite-stage algebras are surjective: the generators at the smaller stage are restrictions of the same generators at the larger stage. Compactness and Hausdorffness follow from (1.1). Reducedness follows from the injections (2.1). Those injections also show that the intersection of all classical kernels in A is zero, which is equivalent to the asserted density. Finally, the $\mathbb{Z}_2$ -algebra of polynomials in the indicated generators surjects onto each A_k , and is therefore dense in the inverse limit. $\square$

For every classical system, the theorems of Deligne and Deligne–Serre [6, 7] give a continuous two-dimensional representation of G over a finite extension of $\mathbb{Q}_2$; in the Eisenstein case one uses a direct sum of characters. Choose the Frobenius convention so that its characteristic polynomial at $\ell \nmid 2N$ is

$$Z^2 - T_\ell Z + \ell S_\ell \quad (2.2)$$

after specialization. A stable lattice ensures integral traces and determinants at every element of G .

2.2 An integral determinant law

A two-dimensional *determinant* over a commutative ring R is a unital multiplicative homogeneous polynomial law of degree two $R[G] \rightarrow R$, compatible with extension of scalars; see [4, Section 1].

For a determinant D , its trace and determinant on a group element g are defined by $D(Z - g) = Z^2 - t(g)Z + d(g)$. This notion works over rings in which 2 is not invertible.

Lemma 2.2. *There is a continuous two-dimensional determinant over A whose trace and determinant satisfy*

$$t(\text{Frob}_\ell) = T_\ell, \quad d(\text{Frob}_\ell) = \ell S_\ell \quad (\ell \nmid 2N).$$

Proof. At a fixed finite stage, take the tuple of traces and determinants of its classical Galois representations. The tuple belongs to the image of (2.1) on Frobenius conjugacy classes, by (2.2). The union of these classes is dense in G by Chebotarev. The tuples are continuous class functions, and the image of (2.1) is closed. Hence the tuples belong to A_k at every $g \in G$. Uniqueness on the dense union of Frobenius classes makes these functions compatible as k varies. They give continuous functions t, d with values in A .

For completeness, the determinant law itself can be constructed without recovering it by division by 2. On a finite formal sum $\sum_g x_g g$, prescribe the quadratic polynomial

$$\sum_g d(g) x_g^2 + \sum_{g < h} (t(g)t(h) - t(gh)) x_g x_h, \quad (2.3)$$

where any ordering of the finite support may be used. In every classical representation this is the determinant of $\sum_g x_g r(g)$; the formula is symmetric since $t(gh) = t(hg)$. All polynomial identities expressing normalization and multiplicativity hold coefficientwise in every classical specialization. Their coefficients therefore vanish in A by Lemma 2.1. Formula (2.3) defines the required polynomial law over arbitrary A -algebras, with trace t and determinant d . $\square$

2.3 Noetherianity at two

Proposition 2.3. *The ring A is a finite product of complete Noetherian local rings with finite residue fields. Its given topology on each factor is the maximal-ideal-adic topology.*

Proof. Jochowitz's finiteness theorem [11], in the form [8, Proposition 2.8], says that the reductions of the classical systems at fixed prime-to-2 level form a finite set. Every maximal ideal of A_k is detected by such a reduction: the product in (2.1) is finite integral over A_k , so one can apply lying over. Each A_k is a finite product of complete local rings with finite residue fields. A surjective transition map $A_{k+1} \rightarrow A_k$ injects the set of maximal ideals of A_k into that of A_{k+1} . These increasing sets have bounded cardinality, so stabilize on a tail. The corresponding local-factor maps are surjective, and identify their residue fields. We obtain

$$A = \prod_{\alpha=1}^s A^{(\alpha)},$$

where each $A^{(\alpha)}$ is a profinite local ring with finite residue field k_α .

The canonical Witt coefficient maps $W(k_\alpha)$ into the finite-stage local rings are the unique unramified lifts of the residue-field identifications. Their uniqueness makes them compatible, and gives $A^{(\alpha)}$ its $W(k_\alpha)$ -algebra structure. Project the determinant of Lemma 2.2 to this factor, and let $\overline{D}_\alpha$ be its reduction. Chenevier's determinant deformation theorem [4, Propositions 3.3 and 3.7] provides a complete Noetherian local universal ring $R_{\overline{D}_\alpha}$ with finite residue field, representing continuous deformations of $\overline{D}_\alpha$. Here the relevant finiteness hypothesis is Mazur's condition: every open subgroup $H \subseteq G$ has only finitely many continuous homomorphisms to $\mathbb{Z}/2\mathbb{Z}$. For

$G_{\mathbb{Q},S}$ this follows from number-field finiteness for quadratic extensions with ramification restricted to a fixed finite set; see also [4, Example 3.6].

Apply the universal property to finite quotients and pass to the inverse limit. This gives a continuous map

$$R_{\overline{D}_\alpha} \longrightarrow A^{(\alpha)}.$$

Its image contains the projections of every T_ℓ and ℓS_ℓ . The image is compact and therefore closed, so Lemma 2.1 makes the map surjective. Its kernel is closed; the quotient is a complete Noetherian local ring. The continuous bijection from this compact quotient to the Hausdorff ring $A^{(\alpha)}$ is a homeomorphism. This proves the topology assertion as well. $\square$

The determinant argument supplies the $p = 2$ justification for the pseudodeformation step in [8, Theorem 2.7], whose original references included an odd-prime restriction. No division by 2 has been used in Proposition 2.3.

We now record the geometric input from the infinite fern.

Theorem 2.4 (Emerton). *For the algebra A in (1.1), every irreducible component of $\mathrm{Spec} A$ has dimension at least 4.*

This is [8, Corollary 2.28], with the definitions in Sections 2.1 and 2.5 of that paper and with $p = 2$. It concerns the Hecke algebra of all modular forms used here, rather than a universal deformation ring for an irreducible residual representation.

3 Finding cuspidal points on every component

The lower bound in Theorem 2.4 lets us exclude small classical loci. We show that Eisenstein systems cannot fill a component, even when all weights are allowed.

Proposition 3.1. *The closure of the classical points of bounded weight has dimension at most 1. The closure of all Eisenstein points has dimension at most 2.*

Proof. Systems of weights at most k factor through A_k , which is finite over $\mathbb{Z}_2$. Their closure therefore lies in the one-dimensional closed subset defined by $\ker(A \rightarrow A_k)$. In particular, we may discard weights one and two when considering the Eisenstein locus.

In weight $i \geq 3$, an Eisenstein system has eigenvalues

$$T_\ell \longmapsto \psi(\ell) + \phi(\ell)\ell^{i-1}, \quad \ell S_\ell \longmapsto \psi(\ell)\phi(\ell)\ell^{i-1}, \quad (3.1)$$

where ψ, ϕ are Dirichlet characters of conductors dividing N . The standard Eisenstein description gives these formulas; oldforms do not change the good-prime system. There are finitely many possible pairs, including their 2-adic embeddings. Partition the systems by pair and by the parity of i . A family with finitely many weights has already been dealt with, so fix an infinite family, indexed by a set I of weights, and a finite extension $E/\mathbb{Q}_2$ containing the character values. Put $\mathcal{O} = \mathcal{O}_E$.

Every odd prime has a unique expression

$$\ell = s_\ell 5^{b_\ell}, \quad s_\ell \in \{1, -1\}, \quad b_\ell \in \mathbb{Z}_2.$$

On the fixed parity class, s_ℓ^{i-1} is constant. In (3.1), replace ℓ^{i-1} by $s_\ell^{i-1}(1+X)^{b_\ell} \in \mathcal{O}[[X]]$. The resulting series specialize correctly at

$$X = x_i := 5^{i-1} - 1 \in 4\mathbb{Z}_2, \quad i \in I. \quad (3.2)$$

We next verify that this prescription extends from generators to the inverse-limit algebra.

Give $\mathcal{O}[[X]]$ its maximal-ideal-adic topology. Let C be the closure in $A \times \mathcal{O}[[X]]$ of the $\mathbb{Z}_2$ -algebra generated by the paired Hecke generators and their prescribed series. The projection $C \rightarrow A$ is surjective: its image is closed by compactness and contains a dense subalgebra. For every $(a, F) \in C$ and $i \in I$, continuity gives

$$\lambda_i(a) = F(x_i), \quad (3.3)$$

where λ_i is the corresponding classical system. If $(0, F) \in C$, then F vanishes at all the distinct x_i . The series $F(4Y)$ is a restricted power series, since its coefficients tend to zero. Strassmann's theorem therefore implies $F(4Y) = 0$, and hence $F = 0$. Thus $C \rightarrow A$ is also injective. It is a homeomorphism, and the second projection defines a continuous map

$$\Phi : A \longrightarrow \mathcal{O}[[X]]. \quad (3.4)$$

Let $B = \Phi(A)$, a compact, hence closed, subring of $\mathcal{O}[[X]]$. Choose a prime $\ell \nmid 2N$ with

$$\ell \equiv 1 \pmod{N}, \quad \ell \equiv 5 \pmod{8}.$$

Such primes exist by the Chinese remainder theorem and Dirichlet's theorem. For this prime $\psi(\ell) = \phi(\ell) = 1$, $s_\ell = 1$, and $b_\ell \in \mathbb{Z}_2^\times$. It follows from the image of ℓS_ℓ that $u = (1 + X)^{b_\ell} \in B$. The series

$$u^{b_\ell^{-1}} = \sum_{n \geq 0} \binom{b_\ell^{-1}}{n} (u - 1)^n = 1 + X$$

converges in $\mathcal{O}[[X]]$, and its partial sums belong to B . Closedness gives $X \in B$ and then $\mathbb{Z}_2[[X]] \subseteq B$. A finite $\mathbb{Z}_2$ -basis of $\mathcal{O}$ consequently generates $\mathcal{O}[[X]]$ as a B -module. This is a finite integral extension, so $\dim B = \dim \mathcal{O}[[X]] = 2$. All the points in this family factor through $B = A / \ker \Phi$ by (3.3); their closure has dimension at most 2. Taking the finite union over the families, and adding weights one and two, proves the claim. $\square$

Corollary 3.2. *For every minimal prime P of A , there is a classical cuspidal map $\lambda : A \rightarrow \mathcal{O}_E$ of weight at least 3 such that $P \subseteq \mathfrak{x} := \ker \lambda$ and $\mathfrak{x}$ belongs to no other irreducible component of $\text{Spec } A$.*

Proof. Let Z be the closure of the Eisenstein points and of the points of weights one and two. By Proposition 3.1, $\dim Z \leq 2$, whereas $\dim V(P) \geq 4$ by Theorem 2.4. Since A is Noetherian, it has finitely many minimal primes. Removing Z and every component other than $V(P)$ leaves an open subset of $\text{Spec } A$ containing the point P . This open subset is nonempty and lies in $V(P)$. Lemma 2.1 supplies a classical point in it, which has the asserted properties. $\square$

Fix a point furnished by Corollary 3.2. Its system comes from a cuspidal newform f of weight $k \geq 3$ and level dividing N . Enlarging E if necessary, let

$$r : G \longrightarrow \text{GL}_2(E) \quad (3.5)$$

be its absolutely irreducible Galois representation. Absolute irreducibility here is a characteristic-zero theorem for cuspidal newforms [17, Theorem 2.3]; nothing is asserted about its reduction. The specializations $\lambda(t), \lambda(d)$ are the trace and determinant of r .

4 A tangent bound in characteristic zero

We prove the tangent estimate in a form that separates it from the particular Hecke algebra. For a representation r , the notation $\mathrm{ad} r$ means $\mathrm{End}_E(E^2)$ with the conjugation action, not its trace-zero subspace.

Proposition 4.1. *Let R be a compact Hausdorff Noetherian topological $\mathbb{Z}_p$ -algebra, with continuous structure map $\mathbb{Z}_p \rightarrow R$. Suppose a profinite group H has a continuous two-dimensional determinant over R , with trace t and determinant d , and that their values topologically generate R over $\mathbb{Z}_p$. Let $\lambda : R \rightarrow \mathcal{O}_E$ be a continuous $\mathbb{Z}_p$ -algebra map, where $E/\mathbb{Q}_p$ is finite, and suppose its specialized determinant is that of an absolutely irreducible continuous representation $r : H \rightarrow \mathrm{GL}_2(E)$. For $\mathfrak{r} = \ker \lambda$,*

$$\dim R_{\mathfrak{r}} \leq \mathrm{edim} R_{\mathfrak{r}} \leq \dim_E H^1(H, \mathrm{ad} r). \quad (4.1)$$

Proof. The first inequality is the usual dimension bound for a Noetherian local ring. We prove the second by relating derivations to first-order Galois deformations.

The algebraic tangent space consists of continuous derivations. The image $R/\mathfrak{r}$ is a $\mathbb{Z}_p$ -submodule of $\mathcal{O}_E$, so is finite over $\mathbb{Z}_p$. Its fraction field $E_0 = \mathrm{Frac}(R/\mathfrak{r})$ is the residue field of $R_{\mathfrak{r}}$ and is finite over $\mathbb{Q}_p$. Since $p \notin \mathfrak{r}$, $R_{\mathfrak{r}}$ is a $\mathbb{Q}_p$ -algebra. Put $\mathfrak{m} = \mathfrak{r}R_{\mathfrak{r}}$. Separability of $E_0/\mathbb{Q}_p$ gives a coefficient field in $R_{\mathfrak{r}}/\mathfrak{m}^2$, and every $\mathbb{Q}_p$ -derivation kills this field. It follows that

$$\mathrm{Der}_{\mathbb{Q}_p}(R_{\mathfrak{r}}, E) \simeq \mathrm{Hom}_{E_0}(\mathfrak{m}/\mathfrak{m}^2, E), \quad (4.2)$$

so the left side has E -dimension $\mathrm{edim} R_{\mathfrak{r}}$. The coefficient field can also be obtained by lifting a primitive element of the finite separable extension across the square-zero ideal; its minimal polynomial has invertible derivative.

Let D be such a derivation, restricted to R . It is $\mathbb{Z}_p$ -linear and vanishes on $\mathfrak{r}^2$. Noetherianity makes $\mathfrak{r}/\mathfrak{r}^2$ finite over $R/\mathfrak{r}$, and the exact sequence

$$0 \longrightarrow \mathfrak{r}/\mathfrak{r}^2 \longrightarrow R/\mathfrak{r}^2 \longrightarrow R/\mathfrak{r} \longrightarrow 0$$

shows that $R/\mathfrak{r}^2$ is finite over $\mathbb{Z}_p$. Every finitely generated ideal of R is closed: it is the image of a continuous map $R^m \rightarrow R$, hence compact. The quotient topology on $R/\mathfrak{r}^2$ is its usual finite-module topology, since a continuous surjection from a finite free $\mathbb{Z}_p$ -module to it is a quotient map between compact Hausdorff spaces. Any $\mathbb{Z}_p$ -linear map from a finite $\mathbb{Z}_p$ -module to E is continuous. In particular, $D : R \rightarrow E$ is continuous.

First-order determinants lift to continuous representations. Write $E[\varepsilon]$ for the dual-number E -algebra, with $\varepsilon^2 = 0$. The map $\lambda + \varepsilon D : R \rightarrow E[\varepsilon]$ specializes the determinant to a continuous determinant lifting that of r . Passing to the canonical Cayley–Hamilton quotient of $E[\varepsilon][H]$ (the quotient imposing the characteristic-polynomial identities of this determinant), the absolutely irreducible lifting theorem for determinants [4, Theorem 2.22(i) and Corollary 2.23] gives a representation

$$r_D : H \longrightarrow \mathrm{GL}_2(E[\varepsilon])$$

with this determinant, whose reduction can be identified with r . Its applicability is over the Henselian local ring $E[\varepsilon]$: the residual representation at this point is the split absolutely irreducible representation r over the characteristic-zero field E .

Here continuity of r_D follows from continuity of its trace. Choose $g_1, \dots, g_4 \in H$ such that the matrices $r(g_j)$ form an E -basis of $M_2(E)$; absolute irreducibility guarantees such a choice.

The lifts $r_D(g_j)$ form a basis over $E[\varepsilon]$. The trace pairing on this matrix algebra is nondegenerate, so the coordinates of $r_D(g)$ in that basis are obtained by inverting a fixed Gram matrix and using the four functions

$$g \longmapsto \operatorname{tr}(r_D(g)r_D(g_j)) = \lambda(t(gg_j)) + \varepsilon D(t(gg_j)).$$

These functions are continuous. Hence r_D is continuous.

Traces detect tangent vectors. Write $r_D(g) = (1 + \varepsilon c(g))r(g)$. The homomorphism identity is exactly the cocycle identity for a continuous $c \in Z^1(H, \operatorname{ad} r)$. Comparing traces gives

$$D(t(g)) = \operatorname{tr}(c(g)r(g)).$$

Coboundaries have zero image under the linear map

$$H^1(H, \operatorname{ad} r) \longrightarrow \operatorname{Maps}(H, E), \quad [c] \longmapsto (g \mapsto \operatorname{tr}(c(g)r(g))), \quad (4.3)$$

where $\operatorname{Maps}(H, E)$ is the E -vector space of all functions $H \rightarrow E$. Thus the image of the linear map $D \mapsto D \circ t$ lies in the image of (4.3).

This map on derivations is injective. Indeed the degree-two trace identity gives

$$2d(g) = t(g)^2 - t(g^2).$$

If $D(t(g)) = 0$ for every g , applying D and working in the characteristic-zero field E gives $D(d(g)) = 0$ for every g . The derivation then vanishes on the algebra of generators, and on its closure by continuity. Its restriction to R determines it on $R_{\mathfrak{r}}$ by the quotient rule. The injection and (4.2) now prove (4.1). $\square$

Apply Proposition 4.1 to $R = A$, $p = 2$, $H = G$, and the classical point chosen above. Propositions and lemmas in Section 2 verify its hypotheses. With $V = \operatorname{ad} r$, we obtain

$$\dim A_{\mathfrak{r}} \leq \operatorname{edim} A_{\mathfrak{r}} \leq \dim_E H^1(G, V). \quad (4.4)$$

The use of characteristic zero in the last proof does not impose any condition on the residual system of the component.

5 The adjoint cohomology bound

We now show that the right side of (4.4) is at most 3. For a finite place v , write $H^j(\mathbb{Q}_v, V)$ for the cohomology of its absolute Galois group. The Bloch–Kato finite subspace is

$$H_f^1(\mathbb{Q}_v, V) = \begin{cases} \ker(H^1(\mathbb{Q}_v, V) \rightarrow H^1(I_v, V)), & v \neq 2, \\ \ker(H^1(\mathbb{Q}_2, V) \rightarrow H^1(\mathbb{Q}_2, V \otimes_{\mathbb{Q}_2} B_{\operatorname{cris}})), & v = 2, \end{cases}$$

where I_v is inertia and B_{cris} is Fontaine’s crystalline period ring. The global group $H_f^1(\mathbb{Q}, V)$ consists of classes satisfying these conditions at every finite place. Conditions at infinity impose nothing here: the positive-degree cohomology of the real Galois group with E -coefficients vanishes, since 2 is invertible in E .

5.1 Global Selmer vanishing

The input from [16, Theorem 5.4], specialized to $\mathrm{GL}_2/\mathbb{Q}$, is the following: if a regular algebraic cuspidal automorphic representation is non-CM, or is CM by a quadratic field not contained in $\mathbb{Q}(\zeta_{2^\infty})$, then the finite Selmer group of its full 2-adic adjoint representation vanishes. There is no hypothesis of residual irreducibility or an odd-prime restriction in this statement.

Lemma 5.1. *For the representation (3.5), $H_f^1(\mathbb{Q}, \mathrm{ad} r) = 0$.*

Proof. The form f has weight at least 3, so its associated cuspidal automorphic representation is regular algebraic, in the usual cohomological normalization. Twists introduced by normalization do not change its adjoint representation.

Only the CM case requires a check. If f has CM by K , its automorphic representation has the quadratic self-twist $\chi_{K/\mathbb{Q}}$. Since the level of f divides the odd integer N , its local representation at 2 is unramified. The local Langlands parameter of an unramified representation has trivial inertia; tensoring it with a ramified character cannot leave it unchanged. Thus $\chi_{K/\mathbb{Q}}$ is unramified at 2. The three quadratic subfields of $\mathbb{Q}(\zeta_{2^\infty})$ are $\mathbb{Q}(\sqrt{-1})$, $\mathbb{Q}(\sqrt{2})$, and $\mathbb{Q}(\sqrt{-2})$, all ramified at 2. Therefore K is not contained in that cyclotomic extension. Newton–Thorne’s theorem applies in both cases and gives the result. We may enlarge the finite coefficient field to apply the theorem; vanishing descends to E under finite extension of scalars. $\square$

5.2 The local quotients

Global Selmer vanishing will make restriction to the sum of the local quotients injective. To compute these quotients, we first use local Tate duality to eliminate the H^2 terms from the local Euler characteristic formulas.

Lemma 5.2. *For every finite place v , $H^0(\mathbb{Q}_v, V^*(1)) = 0$, where $V = \mathrm{ad} r$.*

Proof. The trace pairing identifies V^* with V . The asserted group is therefore $\mathrm{Hom}_{G_{\mathbb{Q}_v}}(r, r(1))$.

At $v = 2$, the representation r is de Rham, hence Hodge–Tate, with two weights whose difference is $k - 1 \geq 2$. These are the classical comparison and filtration statements for modular Galois representations; see [12, Section 11.3, especially (11.3.3)–(11.3.4)]. One may pass to an auxiliary fine level when applying that cohomological description; the representation attached to f is unchanged. A Tate twist shifts both weights by one, so the weight sets of r and $r(1)$ are disjoint. Subrepresentations and quotients of Hodge–Tate representations are Hodge–Tate, with their weights among those of the original representation. A nonzero image of a map $r \rightarrow r(1)$ would have weights in both sets, which is impossible.

Let $v \neq 2$. Local–global compatibility for cuspidal newforms [3, Theorem A] identifies the Frobenius-semisimplified Weil–Deligne parameter of $r|_{G_{\mathbb{Q}_v}}$ with that of the local automorphic representation, up to the conventional normalizing twist. The local automorphic representation is generic. The genericity criterion [1, Lemma 1.1.3] says that its Frobenius-semisimple Weil–Deligne parameter W has $\mathrm{Hom}_{\mathrm{WD}}(W, W(1)) = 0$. Here the Tate twist is normalized consistently with geometric Frobenius; normalizing character twists do not affect the criterion.

A nonzero Galois morphism $r \rightarrow r(1)$ would give a nonzero morphism of Weil–Deligne representations. It would remain a morphism after Frobenius semisimplification: an intertwiner of Frobenius also intertwines its semisimple part, and the inertia and monodromy conditions are unchanged. This contradicts the genericity criterion. $\square$

Proposition 5.3. *For the representation r of a cuspidal newform of weight at least 3 and odd level dividing N ,*

$$\dim_E H^1(G_{\mathbb{Q},S}, \text{ad } r) \leq 3.$$

Proof. Write h^j for the E -dimension of a cohomology group. Local Tate duality and the local Euler characteristic formulas are used in their characteristic-zero form; see [15, Chapter 7]. Duality and Lemma 5.2 give $h^2(\mathbb{Q}_v, V) = 0$ at every finite place. For $v \neq 2$, the local Euler characteristic formula yields $h^1(\mathbb{Q}_v, V) = h^0(\mathbb{Q}_v, V)$. The unramified cohomology is

$$H_f^1(\mathbb{Q}_v, V) = V^{I_v} / (\text{Frob}_v - 1)V^{I_v}.$$

The cokernel and kernel of an endomorphism of a finite-dimensional vector space have the same dimension. Consequently this group has dimension $h^0(\mathbb{Q}_v, V)$, and

$$H^1(\mathbb{Q}_v, V) / H_f^1(\mathbb{Q}_v, V) = 0 \quad (v \neq 2). \quad (5.1)$$

At 2, the Euler characteristic formula instead gives

$$h^1(\mathbb{Q}_2, V) = h^0(\mathbb{Q}_2, V) + \dim_E V = h^0(\mathbb{Q}_2, V) + 4.$$

Write $D_{\text{dR}}(V) = (V \otimes_{\mathbb{Q}_2} B_{\text{dR}})^{G_{\mathbb{Q}_2}}$ for the filtered de Rham module, where B_{dR} is Fontaine's de Rham period field. The Bloch–Kato dimension formula [2, Corollary 3.8.4] gives

$$\begin{aligned} h_f^1(\mathbb{Q}_2, V) &= h^0(\mathbb{Q}_2, V) + \dim_E (D_{\text{dR}}(V) / \text{Fil}^0 D_{\text{dR}}(V)) \\ &= h^0(\mathbb{Q}_2, V) + 1. \end{aligned}$$

Indeed the filtration degrees of the adjoint consist of two zeros and two nonzero opposite integers, so exactly one degree contributes to the indicated quotient. We have proved

$$\dim_E H^1(\mathbb{Q}_2, V) / H_f^1(\mathbb{Q}_2, V) = 3. \quad (5.2)$$

Finally consider restriction followed by passage to these local quotients:

$$H^1(G_{\mathbb{Q},S}, V) \longrightarrow \bigoplus_{v \in S} H^1(\mathbb{Q}_v, V) / H_f^1(\mathbb{Q}_v, V). \quad (5.3)$$

Inflation identifies the source with a subspace of $H^1(G_{\mathbb{Q}}, V)$. Its classes are already unramified outside S , so the kernel of (5.3) lies in $H_f^1(\mathbb{Q}, V)$, which vanishes by Lemma 5.1. Thus (5.3) is injective. Equations (5.1) and (5.2) give a target of dimension 3. $\square$

6 Dimension of the components

Proof of Theorem 1.1. Let P be an arbitrary minimal prime of A . By Corollary 3.2, choose a classical cuspidal point $\mathfrak{r} = \ker \lambda$ of weight at least 3 with $P \subseteq \mathfrak{r}$. Equation (4.4) and Proposition 5.3 give

$$\dim A_{\mathfrak{r}} \leq 3.$$

The quotient A/P is a complete Noetherian local domain by Proposition 2.3, and is therefore catenary. The classical quotient $A/\mathfrak{r}$ is finite over $\mathbb{Z}_2$ and contains $\mathbb{Z}_2$, so has dimension one. The dimension formula for a catenary local domain yields

$$\dim A/P = \text{ht}(\mathfrak{r}/P) + \dim A/\mathfrak{r} \leq \dim A_{\mathfrak{r}} + 1 \leq 4.$$

Theorem 2.4 gives the opposite inequality. Since P was arbitrary, all irreducible components have dimension 4. $\square$

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